

Industrial Internship Report on ”IoT Smart Environment Monitoring System”

Prepared by

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Executive Summary

This report presents the work carried out during the Industrial Internship provided by UpSkill Campus and The IoT Academy in collaboration with industrial partner UniConverge Technologies Pvt. Ltd. (UCT).

The internship focused on developing practical knowledge in Embedded Systems and the Internet of Things (IoT) through learning activities and project implementation.

My project, “**IoT Smart Environment Monitoring System,**” is designed to continuously monitor environmental parameters such as temperature, humidity, and air quality. The proposed system uses an ESP32 microcontroller connected to DHT22 and MQ-2 sensors. The collected sensor information is processed by the ESP32 and can be transmitted through Wi-Fi to an IoT cloud platform for remote monitoring. An LED/buzzer alert mechanism is also incorporated to indicate abnormal environmental conditions.

The project was designed and tested using the Wokwi simulation platform before final documentation. This internship provided practical exposure to IoT architecture, sensors, embedded programming, wireless communication, cloud monitoring, and system testing.

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1 Preface

The Industrial Internship provided an opportunity to understand the practical implementation of Embedded Systems and IoT technologies. Throughout the internship, I studied IoT fundamentals, IoT architecture, IoT devices and platforms, applications, deployment challenges, embedded system concepts, and circuit design.

The internship helped bridge the gap between theoretical concepts and practical engineering applications.

As part of the internship, I selected the project **“IoT Smart Environment Monitoring System.”** The objective of the project is to develop a low-cost monitoring system capable of measuring environmental conditions such as temperature, humidity, and air quality.

The system uses an ESP32 as the main controller, a DHT22 sensor for temperature and humidity measurement, and an MQ-2 sensor for air-quality/gas indication. Wi-Fi connectivity enables environmental information to be transmitted to an IoT platform for remote visualization.

The internship activities followed a systematic approach involving understanding IoT concepts, identifying a practical problem, designing a solution, implementing the proposed system, testing its operation, and documenting the results.

Through this internship, I improved my understanding of IoT, embedded programming, sensor interfacing, circuit design, cloud communication, troubleshooting, and technical documentation.

I express my sincere gratitude to **UpSkill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd., the internship mentors, and my college faculty members** for providing the resources and guidance required to complete this internship.

I would encourage my juniors and peers to participate in practical internships and develop small real-world projects because hands-on implementation greatly improves the understanding of engineering concepts.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



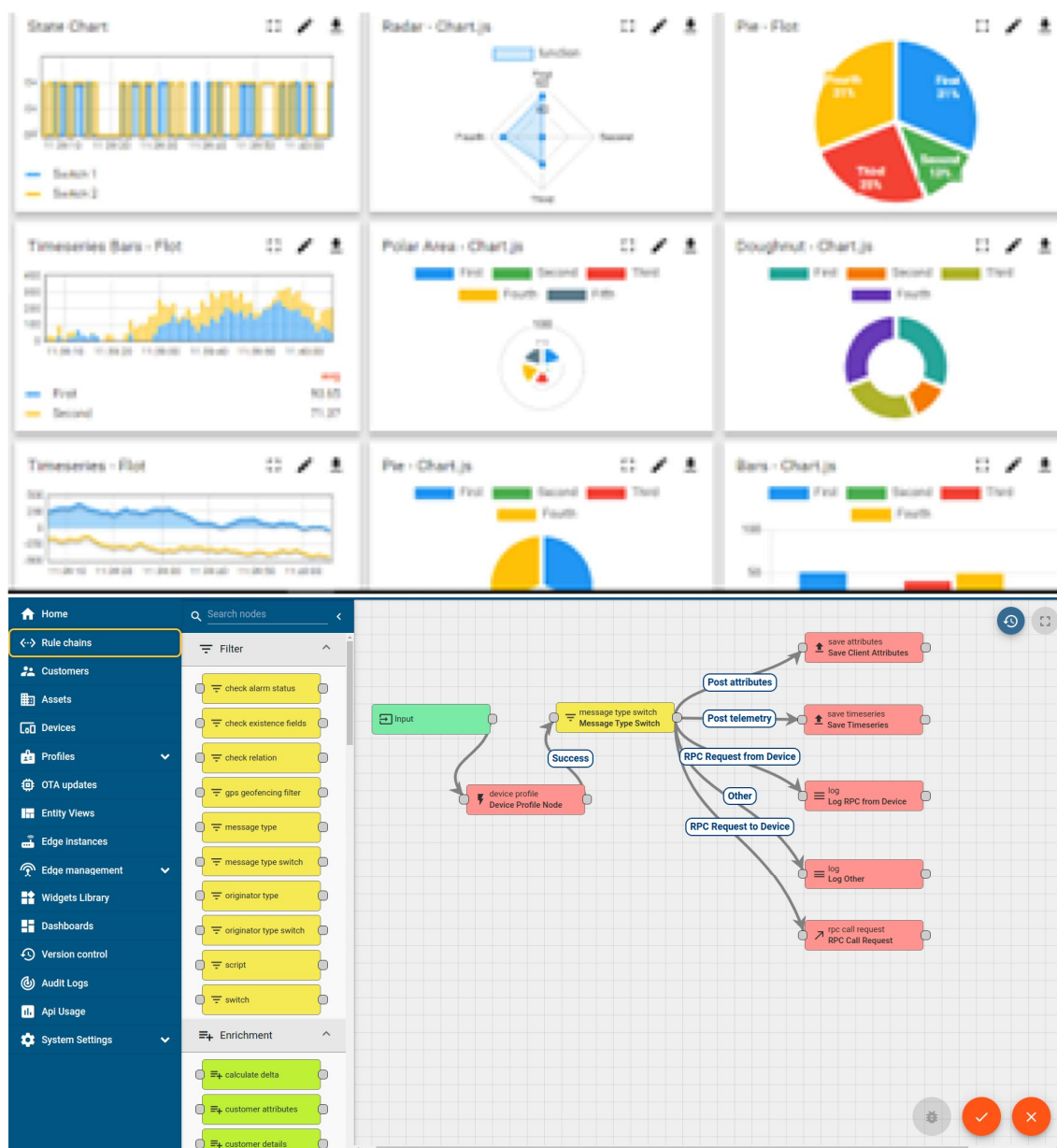
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands. Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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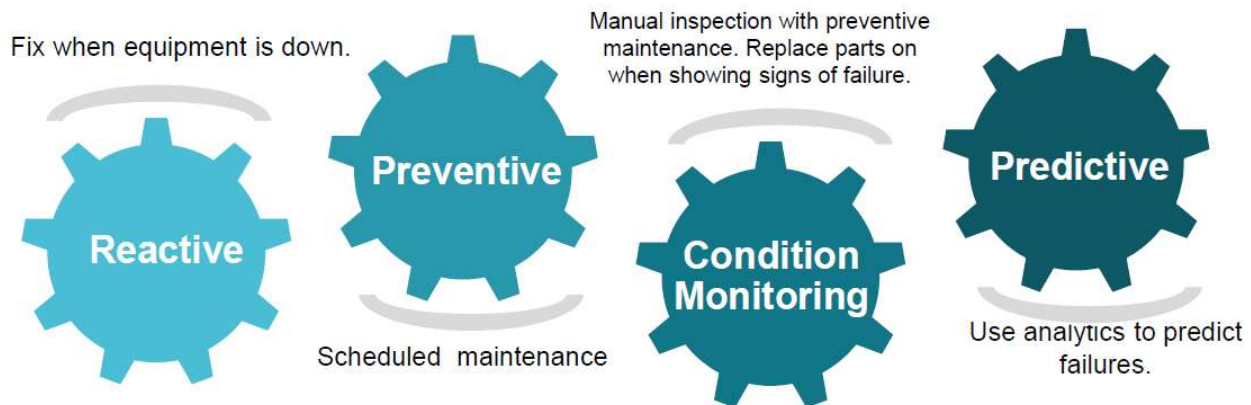


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

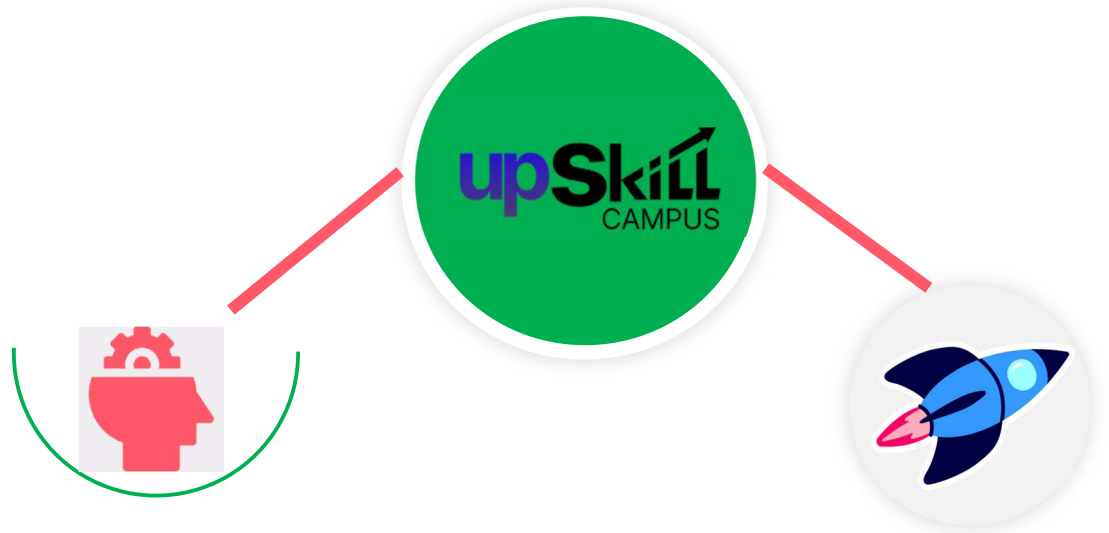
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

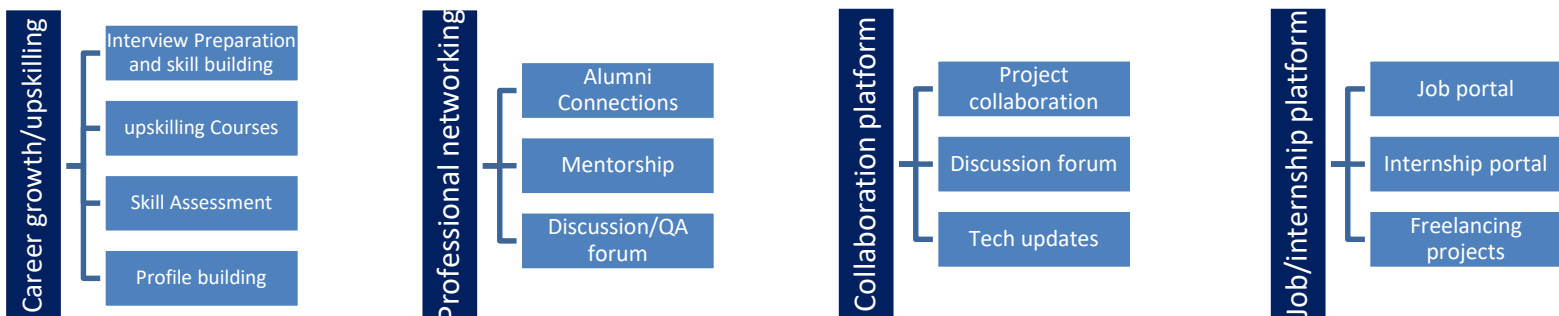
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 Objective

The major objectives were:

- To gain practical experience with Embedded Systems and IoT.
- To understand real-world IoT applications.
- To learn sensor interfacing and microcontroller programming.
- To understand IoT communication and cloud connectivity.
- To improve problem-solving and system-design skills.
- To design and test an IoT-based environmental monitoring application.

2.4 Reference

1. ESP32 Technical Documentation
2. DHT22 Sensor Datasheet
3. MQ-2 Gas Sensor Datasheet
4. Blynk IoT Documentation
5. Wokwi Documentation
6. UpSkill Campus Internship Learning Materials

2.5 Glossary

Term	Acronym
Internet of Things	IoT
Microcontroller Unit	MCU
Wireless Fidelity	Wi-Fi
General Purpose Input/Output	GPIO
Analog-to-Digital Converter	ADC
Message Queuing Telemetry Transport	MQTT
Hypertext Transfer Protocol	HTTP
Integrated Development Environment	IDE
Application Programming Interface	API

3 Problem Statement

Environmental conditions directly influence human comfort, safety, health, and the operation of many industrial and residential environments.

Traditional environmental monitoring may require manual observation or independent instruments. Such approaches may not provide continuous monitoring, centralized visualization, or immediate warnings when environmental conditions become abnormal.

The problem addressed in this project is therefore:

To design a low-cost IoT-based system capable of continuously monitoring temperature, humidity, and gas/smoke conditions and providing remote monitoring and warning functionality.

The system should:

- Continuously measure temperature and humidity.
- Detect changes in gas/smoke sensor output.
- Process sensor readings using a microcontroller.
- Provide local warning indications when predefined limits are exceeded.
- Provide Wi-Fi connectivity.
- Upload readings to an IoT platform.
- Allow users to monitor environmental conditions remotely.

The proposed system can be useful as a prototype for homes, laboratories, offices, storage areas, classrooms, and industrial environments.

4 Existing and Proposed solution

Existing Solution

Conventional environmental monitoring can involve individual thermometers, humidity meters, gas detectors, or manual inspection.

These approaches can have limitations such as:

- Lack of centralized monitoring.

- Limited remote accessibility.
- Requirement for manual observation.
- Separate devices for different environmental parameters.
- Limited historical data visualization.
- Difficulty obtaining immediate information when the user is away.

- **Proposed Solution**

The proposed **IoT Smart Environment Monitoring System** integrates multiple environmental sensors with an ESP32.

The DHT22 measures temperature and humidity while the MQ-2 provides a gas/smoke sensing signal. The ESP32 periodically collects these readings and evaluates them against predefined thresholds.

Through the ESP32's Wi-Fi capability, environmental readings can also be transmitted to **Blynk IoT Cloud** for visualization and remote observation. **Value Addition**

The proposed solution provides:

- Continuous environmental monitoring.
- Multiple sensor parameters in a single system.
- Wireless connectivity.
- Remote visualization.
- Local warning indication.
- Low-cost implementation.
- Easy modification and scalability.
- Simulation-based verification before hardware implementation.

4.1 Code submission (Github link)

<https://github.com/SanjayK-46/upskillCampus>

4.2 Report submission (Github link) :

https://github.com/SanjayK-46/upskillCampus/blob/main/IoT_Smart_Environment_Monitoring_System_SanjayK_USC_UCT.pdf

5 Proposed Design/ Model

The system follows a sensing-processing-communication architecture.

Environmental sensors collect information from the surrounding environment. The ESP32 receives and processes the sensor values. The processed information is displayed through the serial monitor and transmitted through Wi-Fi to the IoT platform.

If a measured parameter crosses its configured threshold, the ESP32 activates the warning mechanism.

5.1 High Level Diagram

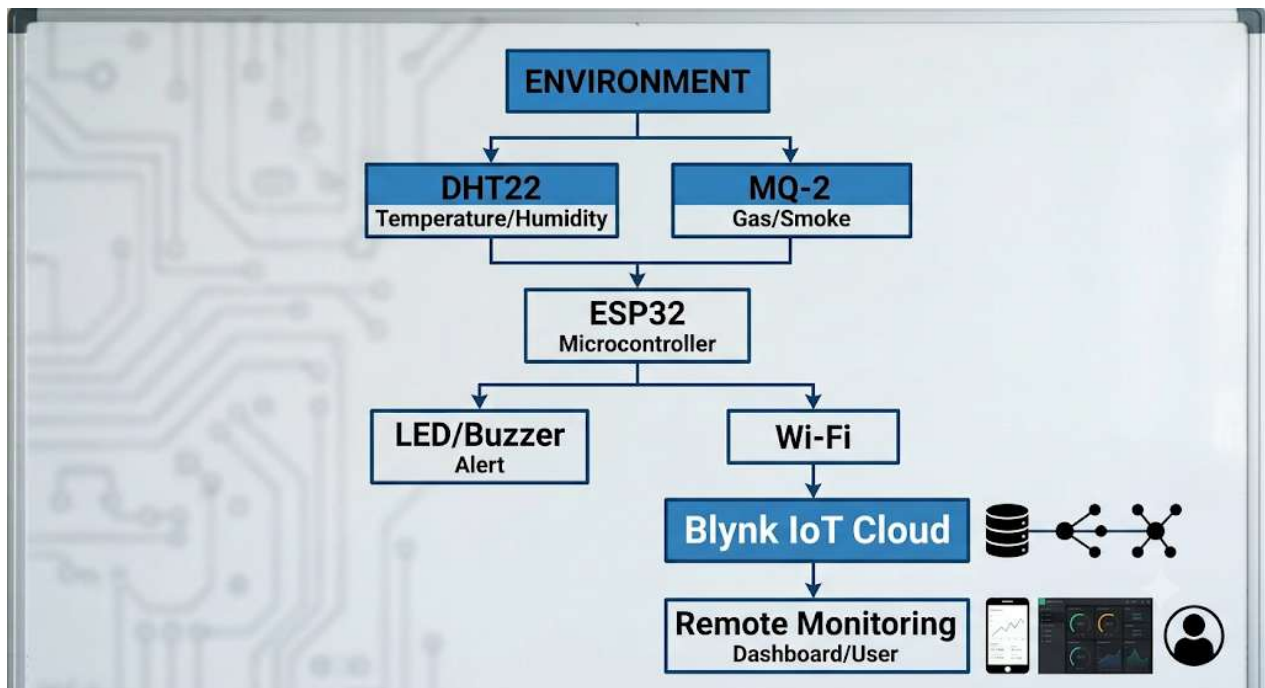


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

The proposed hardware connections are:

DHT22 DATA —————> ESP32 GPIO
DHT22 VCC —————> Supply
DHT22 GND —————> GND

MQ-2 AO —————> ESP32 ADC GPIO
MQ-2 VCC —————> Supply
MQ-2 GND —————> GND

ESP32 GPIO —————> LED
ESP32 GPIO —————> Buzzer

ESP32 Wi-Fi —————> Internet
Internet —————> Blynk IoT Cloud

5.3 Interfaces

➤ Sensor Interface

The DHT22 provides digital temperature and humidity readings to the ESP32.

The MQ-2 sensor provides a signal representing the detected gas/smoke level, which is read through an ESP32 input.

➤ Wi-Fi Interface

The ESP32 connects to a Wi-Fi network and establishes communication with the IoT cloud service.

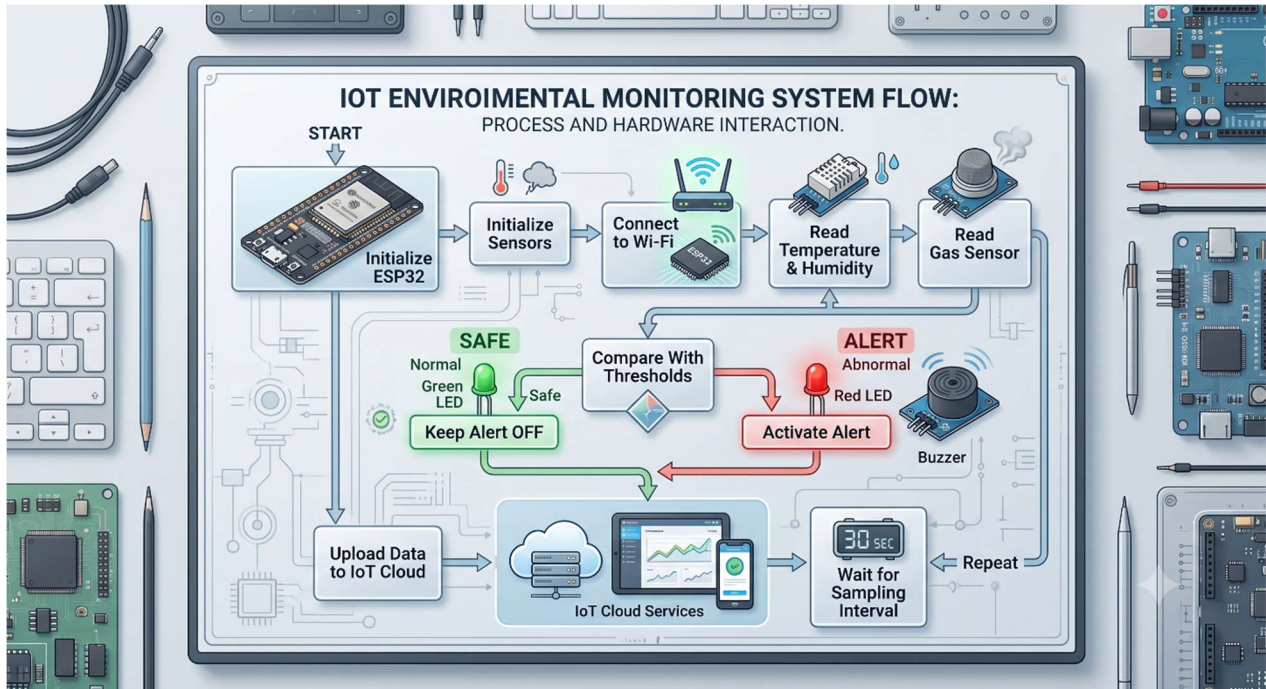
➤ Cloud Interface

Sensor information is transmitted to the Blynk IoT Cloud platform using the ESP32 Wi-Fi module. Users can monitor temperature, humidity, gas level, and system status remotely through the Blynk dashboard.

➤ Alert Interface

When sensor readings exceed defined threshold conditions, the ESP32 activates an LED and/or buzzer to indicate an abnormal condition.

System Flow



6 Performance Test

The template emphasizes that performance testing is important because it demonstrates that the project is more than an academic concept.

For this project, we'll evaluate:

- Correct sensor reading acquisition.
- Response to temperature changes.
- Response to humidity changes.
- Response to simulated gas/smoke changes.
- Alert activation.
- Wi-Fi connectivity.
- Cloud data transmission.
- System stability during repeated monitoring.

6.1 Test Plan/ Test Cases

Test	Input/Condition	Expected Result
TC01	Normal temperature	Temperature displayed correctly
TC02	Temperature above threshold	Warning activated
TC03	Normal humidity	Humidity displayed correctly
TC04	Abnormal humidity	Warning condition detected
TC05	Low MQ-2 value	Normal status
TC06	MQ-2 above configured threshold	Gas/smoke warning activated
TC07	Wi-Fi available	ESP32 connects successfully
TC08	Cloud connection available	Sensor data uploaded
TC09	Multiple sensor updates	System continues monitoring without interruption

6.2 Test Procedure

- Create the ESP32 circuit in Wokwi.
- Connect the DHT22 and MQ-2 sensors.
- Connect the LED/buzzer alert circuit.
- Compile and start the ESP32 program.
- Observe sensor readings through the serial monitor.
- Change DHT22 temperature and humidity values in the simulation.
- Change the simulated MQ-2 sensor condition.
- Verify whether the warning system responds to threshold violations.
- Establish Wi-Fi connectivity.
- Verify cloud data transmission using the **Blynk IoT Cloud**.
- Record screenshots and actual observations for the report.

6.3 Performance Outcome

The **IoT Smart Environment Monitoring System** was successfully designed, simulated, and tested using the Wokwi platform. The ESP32 microcontroller effectively interfaced with the DHT22 sensor, MQ-2 gas sensor, 16×2 I2C LCD, LED, buzzer, and Blynk IoT Cloud platform.

During testing, the system continuously monitored environmental parameters, including **temperature, humidity, and gas concentration**, and displayed the readings on both the LCD and the Blynk dashboard in real time. The ESP32 maintained a stable Wi-Fi connection and successfully transmitted sensor data to the Blynk Cloud without communication issues.

The alert mechanism functioned correctly by activating the **LED and buzzer** whenever the gas sensor value exceeded the predefined threshold. The system status was updated as "**SAFE**" or "**DANGER**" based on the measured gas concentration and was reflected on both the LCD display and the Blynk dashboard.

The project demonstrated reliable performance throughout the simulation. All hardware components responded as expected, and the cloud dashboard accurately displayed real-time sensor values. The implementation successfully achieved the project objectives of **environmental monitoring, remote IoT-based data visualization, and automatic alert generation**, making it suitable as a low-cost prototype for smart homes, laboratories, offices, and industrial environments.

Parameter	Result
Temperature Monitoring	Successfully monitored
Humidity Monitoring	Successfully monitored
Gas Detection	Successfully monitored
LCD Display	Working correctly
LED Alert	Working correctly
Buzzer Alert	Working correctly
Wi-Fi Connectivity	Stable
Blynk Cloud Communication	Successful
Real-Time Dashboard Update	Successful
Overall System Performance	Successful

7 My learnings

The Industrial Internship provided me with valuable practical knowledge in Embedded Systems and the Internet of Things (IoT). During the internship, I learned about IoT architecture, embedded systems, sensor interfacing, cloud connectivity, and wireless communication.

Through the implementation of the **IoT Smart Environment Monitoring System**, I gained hands-on experience in programming the ESP32 microcontroller, interfacing the **DHT22** temperature and humidity

sensor, MQ-2 gas sensor, I2C LCD, LED, and buzzer. I also learned to transmit real-time sensor data to the **Blynk IoT Cloud** for remote monitoring.

The project enhanced my understanding of IoT system design, debugging, simulation using Wokwi, cloud dashboards, and real-time data visualization. It also improved my problem-solving ability, programming skills, and technical documentation.

Overall, this internship strengthened my practical knowledge and increased my confidence in developing real-world IoT applications.

8 Future work scope

The proposed IoT Smart Environment Monitoring System can be further enhanced by adding additional sensors such as smoke, flame, light intensity, and air quality sensors for more comprehensive environmental monitoring.

Future improvements may include:

- Integration with AI-based environmental analysis.
- Mobile notification and SMS alert services.
- Data logging and historical trend analysis.
- GPS-based location tracking.
- Solar-powered operation for outdoor deployment.
- Integration with smart home automation systems.
- Remote device control using the Blynk mobile application.
- Hardware implementation for real-time industrial and residential monitoring.

These enhancements will improve the system's functionality, reliability, and applicability in smart homes, industries, laboratories, and environmental monitoring applications.